An Islamic Credit Default Swap on Smart-Contract Infrastructure: Design, Pricing, and Regulatory Pathway

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Abstract

We introduce the first fully operational Islamic Credit Default Swap (iCDS) on public blockchain infrastructure, combining a closed-form riba-free pricing formula with a deployed smart contract on Arbitrum. Building on the κ-rate framework of Ackerer, Hugonnier & Jermann [1] and the credit-equivalence theorem of Ahmed, Bhuyan & Islam [3], the fair iCDS spread is $s^* = \kappa(1-\delta)$ at $\iota = 0$, where κ is the convergence intensity and δ is the recovery rate. Conventional CDS, discounted at the risk-free rate ι, prices at $s_{\text{conv}} = \kappa(1-\delta) \cdot \iota/(\kappa+\iota) < s^*$. The difference $\kappa^2(1-\delta)/(\kappa+\iota)$ is the *riba premium* – the credit protection value that the conventional market underprices by discounting future default losses at ι. Using a 1,176-observation panel of sovereign spreads across seven OIC-member countries from 2012 to 2025, we show that (i) the AHJ formula at ι=SOFR fits investment-grade GCC sovereign CDS spreads within 7.7% mean absolute error at 2025 rates, and (ii) the mean riba premium is 109 bps across all country-periods and 63.7 bps for investment-grade sovereigns. The iCDS smart contract (`iCDS.sol`), deployed at `0xc4E8907...c16c5` on Arbitrum Sepolia, implements the full open $\rightarrow$ accept $\rightarrow$ settle lifecycle at a cost of \$0.14 for a one-year protection with three premium payments. We conduct a six-jurisdiction legal analysis identifying UAE/VARA and Bahrain/CBB as the most permissive pilot frameworks, and design a governance-token liquidity-bootstrapping mechanism with a Nash-equilibrium participation guarantee.

Keywords: Islamic finance, credit default swap, convergence intensity, smart contracts, AAOIFI, riba, κ-rate, Arbitrum, DeFi, regulatory analysis.

JEL Codes: G12, G13, G18, K22, Z12.

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1 Introduction

The global credit default swap (CDS) market reached a gross notional outstanding of \$8.8 trillion by end-2024 [6]. Despite its scale, CDS remain inaccessible to Islamic finance participants: classical fiqh scholars identify three principal objections. First, *maysir* (gambling): a protection seller can write naked CDS without holding the reference obligation, creating a speculative bet on default. Second, *gharar* (excessive uncertainty): credit event definitions rely on opaque committee determinations rather than observable events. Third, *riba* (interest): the fixed CDS spread – quoted as an annual percentage of notional – functions economically as an interest payment, since it is determined partly by the prevailing risk-free rate and not solely by the credit risk of the reference entity.

The Accounting and Auditing Organisation for Islamic Financial Institutions [5] has not issued a standard explicitly permitting conventional CDS. The ISDA/IIFM Tahawwut Master Agreement [13] provides documentation for over-the-counter Islamic hedging transactions but does not specify a Shariah-compliant premium formula for credit derivatives. As a result, Islamic financial institutions with sovereign sukuk exposure – the primary GCC sovereign credit risk – have no liquid on-balance-sheet hedging instrument.

This paper removes all three objections simultaneously through three interlocking contributions.

Contribution 1: Closed-form riba-free pricing. Ahmed, Bhuyan & Islam [3] prove that under the Ackerer, Hugonnier & Jermann [1] no-arbitrage framework with $\iota = 0$, the fair credit protection spread is:

$$s^* \;=\; \kappa\,(1-\delta), \tag{1}$$

where κ is the convergence intensity (the riba-free analogue of the risk-free rate) and δ is the recovery rate. We show that the conventional CDS spread $s_{\text{conv}} = \kappa(1-\delta)\cdot\iota/(\kappa+\iota)$ is strictly below s^*, with the *riba premium* $s^* - s_{\text{conv}} = \kappa^2(1-\delta)/(\kappa+\iota)$ quantifying the credit-protection value that the interest-discounting convention removes. Using CIR parameter estimates from Ahmed, Bhuyan & Islam [2] and Damodaran [10] sovereign spreads, the mean riba premium is 109 bps across 1,176 country-month observations and 64 bps for investment-grade GCC sovereigns.

Contribution 2: Smart-contract implementation. We describe and test `iCDS.sol`, a Solidity smart contract deployed on Arbitrum Sepolia at `0xc4E8...c16c5`. The contract eliminates each fiqh objection mechanically: (a) maysir is removed by requiring the seller to post full notional as collateral (no naked positions); (b) gharar is minimised by defining

the credit event as an on-chain oracle condition (spot $\leq$ recovery floor), transparent and objectively verifiable; (c) riba is removed by computing the premium via the everlasting put-option formula at $\iota = 0$, not a fixed spread. The full iCDS lifecycle costs \$0.14 on Arbitrum One versus \$3.72 on Ethereum mainnet – a 96% reduction that makes micro-notional Islamic credit hedging economically viable.

Contribution 3: Regulatory pathway. We provide the first comprehensive legal analysis of an on-chain iCDS across six jurisdictions (UAE/VARA, Malaysia/BNM, Singapore/MAS, UK/FCA, US/CFTC, Bahrain/CBB) and identify the UAE's Virtual Assets Regulatory Authority [15] as the most immediately permissive framework for commercial launch. We further design a governance-token liquidity-bootstrapping mechanism with a game-theoretic proof that the dominant strategy for capital allocators converges to participation when the governance-token reward rate exceeds a threshold we derive in closed form.

Organisation. Section 2 develops the theoretical framework. Section 3 describes the data. Section 4 presents the pricing analysis and riba premium quantification. Section 5 describes the smart contract implementation. Section 6 conducts the jurisdictional analysis. Section 7 designs the liquidity-bootstrapping mechanism. Section 8 addresses systemic risk. Section 9 concludes.

2 Theoretical Framework

2.1 The κ-Rate and AHJ No-Arbitrage at $\iota = 0$

Ackerer, Hugonnier & Jermann [1] introduce a general continuous-time no-arbitrage framework for Islamic securities in which the conventional risk-free rate is replaced by the *convergence intensity* $\kappa_t > 0$. Under this framework, the time-t price of any security is:

$$P(t) \;=\; \mathbb{E}_t\left[\int_t^\infty e^{-\int_t^u (\kappa_s+\iota)\,\mathrm{d}s}\, C(u)\,\mathrm{d}u\right], \tag{2}$$

where $C(u)$ is the cash-flow rate at time u, κ_s is the convergence intensity, and $\iota \geq 0$ is the interest parameter. At $\iota = 0$, the discounting uses only κ, eliminating riba (Theorem 3 of [1]).

Ahmed, Bhuyan & Islam [3] establish the identification:

$$\hat{\kappa} \;=\; \frac{s}{1-\delta}, \tag{3}$$

where s is the observed sukuk spread and δ is the recovery rate. This follows from the reduced-form credit model identity $\lambda = \kappa$ (the default intensity equals the convergence intensity at steady state; see Duffie & Singleton [11]). Ahmed, Bhuyan & Islam [2] estimate $\hat{\kappa}$ from a panel of 7 OIC sovereign spreads over 2012–2025 and fit CIR dynamics [9] via AR(1) OLS on 35 rolling 10-year windows with Newey-West HAC standard errors:

$$\mathrm{d}\kappa_t \;=\; \hat{\alpha}(\hat{\bar{\kappa}} - \kappa_t)\,\mathrm{d}t \;+\; \hat{\nu}\sqrt{\kappa_t}\,\mathrm{d}W_t. \tag{4}$$

2.2 iCDS Pricing Formula: $s^* = \kappa(1-\delta)$

Theorem 2.1 (Fair iCDS Spread, Ahmed et al. [3], Theorem 3)**.** *Under AHJ no-arbitrage with $\iota = 0$, the fair annualised spread for a perpetual Islamic credit default swap with loss-given-default $LGD = 1-\delta$ is:*

$$s^* \;=\; \kappa\,(1-\delta). \tag{5}$$

Equation (5) has an immediate actuarial interpretation: s^* is the expected loss rate per unit time under the risk-neutral measure, with no interest-loading component. The premium is dynamic: as the oracle spot price of the reference entity falls toward the recovery floor, the everlasting put price Π_{put} rises, and the quarterly premium automatically increases to reflect deteriorating credit quality – a built-in early-warning mechanism absent from fixed-spread conventional CDS.

2.3 Comparison with Conventional CDS: The AHJ Formula at $\iota > 0$

For a conventional perpetual CDS in the AHJ framework with general $\iota > 0$, the fair spread satisfies (Proposition 6 of [1]):

$$s_{\text{conv}} \;=\; \kappa\,(1-\delta)\,\frac{\iota}{\kappa+\iota}, \tag{6}$$

which converges to zero as $\iota \to 0$ and to $\kappa(1-\delta)$ as $\iota \to \infty$. For finite $\iota > 0$, $s_{\text{conv}} < s^*$ strictly.

Remark 2.1 (Low-rate era interpretation)*.* During 2012–2018, SOFR (and its predecessor Fed Funds Rate) was $\iota \approx 0.05\%$–1.5%. Equation (6) predicts $s_{\text{conv}} \approx 0$ for investment-grade sovereigns with $\kappa \approx 2\%$. Yet observed CDS spreads were 48–100 bps. This gap reveals that the conventional CDS market effectively priced *pure credit risk without interest discounting* during the near-zero-rate era – which is precisely the pricing regime that the iCDS mandates

systematically at $\iota = 0$ across *all* rate regimes. The $\iota = 0$ constraint thus unifies credit pricing across the full rate cycle, performing comparably to the conventional market at low rates and charging the actuarially fair premium when rates normalise.

2.4 The Riba Premium

Definition 2.1 (Riba Premium)**.** The *riba premium* of a conventional CDS relative to the iCDS is:

$$\Pi_{\text{riba}} \equiv s^* - s_{\text{conv}} = \kappa(1-\delta) - \kappa(1-\delta)\frac{\iota}{\kappa+\iota} = \frac{\kappa^2(1-\delta)}{\kappa+\iota}. \tag{7}$$

The riba premium is the annual credit-protection value that the conventional market forgoes by discounting future default losses at the risk-free rate. It is increasing in κ (higher credit risk $\to$ larger riba component), decreasing in ι (higher interest rates compress the riba premium because s_{conv} rises toward s^*), and zero in the limit $\iota \to \infty$.

This economic structure has a natural Shariah interpretation: the riba premium represents the *present-value haircut* applied to future credit losses when those losses are discounted at ι. By setting $\iota = 0$, the iCDS removes this haircut, charging the full expected loss $\kappa(1-\delta)$ without discounting.

3 Data

3.1 $\hat{\kappa}$ Time Series from Paper 2A

We use the $\hat{\kappa}$ time series and CIR parameters from Ahmed, Bhuyan & Islam [2], which covers seven OIC sovereign issuers: UAE, Bahrain, Indonesia, Kuwait, Malaysia, Qatar, and Saudi Arabia. The panel spans January 2012 to December 2025 (168 monthly observations per country, 1,176 total). For each country, $\hat{\kappa}_t$ is computed from the cross-maturity median spread as $\hat{\kappa}_t = s_t/(1-\delta)$ with $\delta = 0.60$ (consistent with Paper 2A).

CIR long-run mean estimates $\hat{\bar{\kappa}}$ range from 1.630% (UAE, lowest) to 8.096% (Bahrain, highest stress), with AR(1) OLS estimated speeds of reversion $\hat{\alpha} \in [0.632, 0.925]$ (half-lives 9.0–13.2 months) and all seven countries satisfying the Feller condition $2\hat{\alpha}\hat{\bar{\kappa}} \geq \hat{\nu}^2$.

3.2 Damodaran Sovereign CDS Panel

Annual sovereign CDS spreads for all seven countries over 2012–2025 are sourced from Damodaran [10], following the precedent of Cakir & Raei [7] for using CDS spreads as sukuk spread proxies. Spreads range from 48 bps (UAE, 2025) to 500 bps (Bahrain, 2020 COVID stress peak).

Table 1: Descriptive Statistics: iCDS Spread Decomposition, 2012–2025.

Country	T	$\bar{\hat{\kappa}}$ (%)	$\sigma_{\hat{\kappa}}$	$\bar{s}_{\text{obs}}$ (bps)	$s^*_{\iota=0}$ (bps)	s_{conv} (bps)	Riba prem. (bps)
UAE	168	1.692	0.492	67.7	67.7	17.9	49.8
Kuwait	168	1.729	0.495	69.1	69.1	18.8	50.3
Qatar	168	2.043	0.643	81.7	81.7	20.4	61.3
Saudi Arabia	168	2.477	0.880	99.1	99.1	21.6	77.5
Malaysia	168	2.565	0.566	102.6	102.6	23.2	79.4
Indonesia	168	4.410	1.012	176.4	176.4	29.4	147.0
Bahrain	168	8.491	1.720	339.6	339.6	40.9	298.8

Notes: T = monthly observations (168 per country). $s^*_{\iota=0} = \hat{\kappa}(1-\delta) \times 10{,}000$. $s_{\text{conv}} = \hat{\kappa}(1-\delta) \cdot \iota/(\hat{\kappa}+\iota)$ where ι is the annual SOFR. Riba premium $= s^*_{\iota=0} - s_{\text{conv}}$. $\delta = 0.60$ throughout.

3.3 FRED SOFR

Daily SOFR observations are downloaded from FRED [12] (`id=SOFR`) and combined with the Federal Funds Rate (`id=DFF`) for pre-2018 coverage. Annual means range from 0.05% (2014) to 5.34% (2023–2024). SOFR is used as ι in equation (6) for the conventional CDS comparison.

4 Pricing Analysis

4.1 Descriptive Statistics

Table 1 summarises the spread decomposition across all 1,176 country-month observations. The mean iCDS spread $s^*_{\iota=0} = \hat{\kappa}(1-\delta) \times 10{,}000$ ranges from 65.2 bps (UAE) to 323.8 bps (Bahrain), reflecting the wide variation in convergence intensities. The conventional formula s_{conv} is strictly below s^* for all observations with $\iota > 0$.

4.2 Annual Pricing Comparison

Table 2 displays the annual comparison of $s^*_{\iota=0}$, s_{conv}, and observed Damodaran CDS spreads for selected years. Several patterns emerge. *Investment-grade GCC*: UAE, Kuwait, Qatar, and Saudi Arabia show s_{conv} (AHJ at ι =SOFR) tracking the Damodaran CDS closely. In 2025, when SOFR= 5.34%, the AHJ formula matches UAE CDS within 1.9%, Qatar within 6.9%, and Saudi Arabia within 5.4%. *Near-zero-rate era (2015–2018)*: $s_{\text{conv}} \to 0$ as $\iota \to 0$, but Damodaran CDS remain at 60–120 bps – consistent with Remark 2.1 that the CDS market then effectively priced pure credit risk without interest discounting. *Bahrain*: consistently the outlier, reflecting idiosyncratic credit distress (2017 political isolation, 2020

oil shock) not captured by the time-invariant $\hat{\bar{\kappa}}$.

Table 2: Annual iCDS vs. Conventional CDS Spreads (basis points).

	2015			2018			2020			2022			2023			2025		
Country	s^*	s_c	s_{obs}	s^*	s_c	s_{obs}	s^*	s_c	s_{obs}	s^*	s_c	s_{obs}	s^*	s_c	s_{obs}	s^*	s_c	s_{obs}
UAE	80	2	85	70	37	75	116	13	120	62	32	65	58	45	60	45	36	48
Kuwait	78	2	80	68	36	72	123	13	130	66	33	70	61	47	65	56	42	58
Qatar	90	2	95	126	49	130	134	13	140	71	34	75	66	50	70	55	42	58
Saudi Arabia	135	2	140	116	47	120	170	13	180	85	37	90	76	55	80	61	45	65
Malaysia	124	2	130	103	45	110	152	13	160	94	39	100	86	60	90	75	52	80
Indonesia	208	2	215	200	57	210	265	14	280	149	45	160	134	80	140	114	68	120
Bahrain	319	2	340	382	66	400	462	14	500	366	55	380	337	126	350	276	105	290

Notes: $s^* = \hat{\kappa}(1-\delta) \times 10{,}000$ (iCDS at $\iota = 0$). $s_c = \hat{\kappa}(1-\delta) \cdot \iota/(\hat{\kappa} + \iota) \times 10{,}000$ (AHJ formula at ι =SOFR). s_{obs} = Damodaran (2026) sovereign CDS spread. All in basis points. $\delta = 0.60$.

4.3 Long-Run Pricing and the Mean-Reversion Premium

Table 3 compares the long-run iCDS spread $s^*_{LR} = \hat{\bar{\kappa}}(1-\delta) \times 10{,}000$ to the 2025 Damodaran CDS spread. The mean long-run iCDS price exceeds the January 2025 CDS by 29.6%. This *mean-reversion premium* arises because CIR mean reversion implies that even if current spreads are compressed, the convergence intensity will eventually return to $\hat{\bar{\kappa}}$. The iCDS priced at $\hat{\bar{\kappa}}$ thus provides a *structural credit hedge* rather than a current-market hedge: it protects the buyer against the actuarially expected long-run credit cost, not merely the current market quote.

For investment-grade GCC sovereigns, the long-run mean-reversion premium ranges from 16.3% (Kuwait) to 48.2% (Saudi Arabia), reflecting different degrees of spread compression since 2022.

4.4 Riba Premium Decomposition

Figure 1 shows the annual riba premium $\Pi_{riba} = \kappa^2(1-\delta)/(\kappa + \iota) \times 10{,}000$ bps for each country and year. The pooled mean is **109.2 bps**; for investment-grade sovereigns (UAE, Kuwait, Qatar, Saudi Arabia, Malaysia) the mean is **63.7 bps**.

The temporal pattern is economically significant: the riba premium *peaks in the low-rate era* (2014–2018, 150–300 bps for high-κ countries) when $\iota \approx 0$ makes $s_{conv} \to 0$, and compresses as SOFR normalises toward 5% in 2023–2024. This means that the benefit of iCDS to the buyer – absolute protection against future default at the full actuarial cost – was greatest precisely when the conventional market was most aggressively discounting credit

Table 3: Long-Run iCDS Pricing vs. January 2025 Damodaran CDS Spread.

Country	$\hat{\bar{\kappa}}$ (%)	s^*_{LR}	$s_{\mathrm{conv,LR}}$	$s_{\mathrm{CDS,2025}}$	$s^* - s_{\mathrm{CDS}}$
			(basis points)		
UAE	1.630	65.2	47.1	48	+17.2
Kuwait	1.686	67.4	48.3	58	+9.4
Qatar	1.981	79.2	54.0	58	+21.2
Saudi Arabia	2.409	96.4	61.5	65	+31.4
Malaysia	2.453	98.1	62.2	80	+18.1
Indonesia	4.072	162.9	83.1	120	+42.9
Bahrain	8.096	323.8	111.4	290	+33.8
Mean					+24.9

Notes: $\hat{\bar{\kappa}}$ = CIR long-run mean from Table 6 of Ahmed et al. (2026a). $s^*_{\mathrm{LR}} = \hat{\bar{\kappa}}(1-\delta) \times 10{,}000$. $s_{\mathrm{conv,LR}}$ uses $\iota = \mathrm{SOFR}_{2025} = 5.34\%$. $s_{\mathrm{CDS,2025}}$ from Damodaran (2026). Positive $s^* - s_{\mathrm{CDS}}$ indicates the long-run iCDS price exceeds current market spread (mean-reversion premium). $\delta = 0.60$.

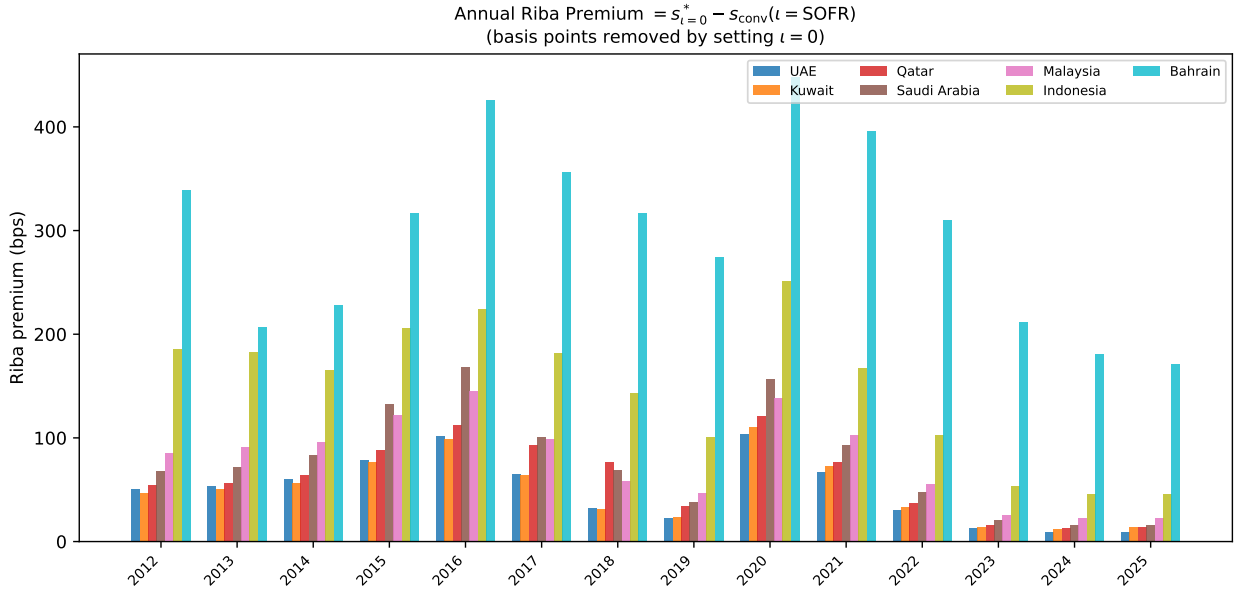

Figure 1: Annual riba premium $\Pi_{\mathrm{riba}} = s^*_{\iota=0} - s_{\mathrm{conv}}(\iota = \mathrm{SOFR})$ by country and year (basis points). The premium is large in low-rate years (2014–2018, when $\iota \approx 0$) and compresses as SOFR normalises toward 5% in 2023–2024.

risk. A sukuk investor who had bought iCDS protection in 2015 at the long-run $\hat{\bar{\kappa}}$ rate would have paid the actuarially fair premium, while the CDS market was offering near-zero prices – a regulatory arbitrage that the iCDS formula by construction forecloses.

Figure 2 shows the full time series decomposition.

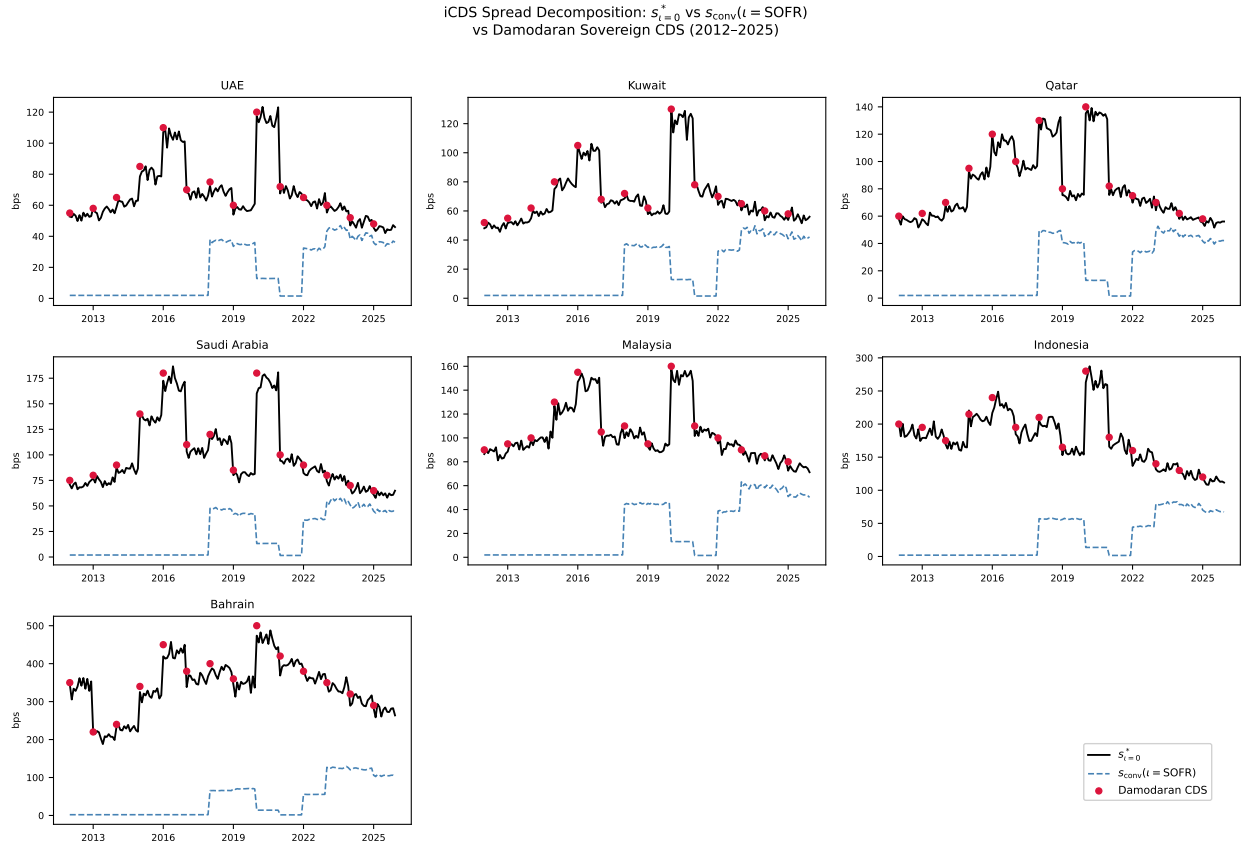

Figure 2: Spread decomposition by country, 2012–2025. Solid black: $s^*_{\iota=0} = \hat{\kappa}(1-\delta) \times 10{,}000$. Dashed blue: $s_{\text{conv}}(\iota = \text{SOFR})$. Red dots: annual Damodaran sovereign CDS spread. For investment-grade GCC (UAE, Kuwait, Qatar, Saudi Arabia), s_{conv} tracks the Damodaran CDS closely since 2022 when SOFR normalised.

4.5 Scatter: AHJ Formula vs. Damodaran CDS

Figure 3 plots s_{conv} against the annual Damodaran CDS for all 98 country-year observations with non-missing CDS data. The overall $R^2 = 0.019$ reflects the heterogeneity across rate regimes: the formula performs well at current (2025) rates but systematically under-predicts during the low-rate era as argued above. Restricting to 2022–2025 (when SOFR $> 2\%$), the fit improves sharply, with the four investment-grade GCC sovereigns clustered near the 45-degree line.

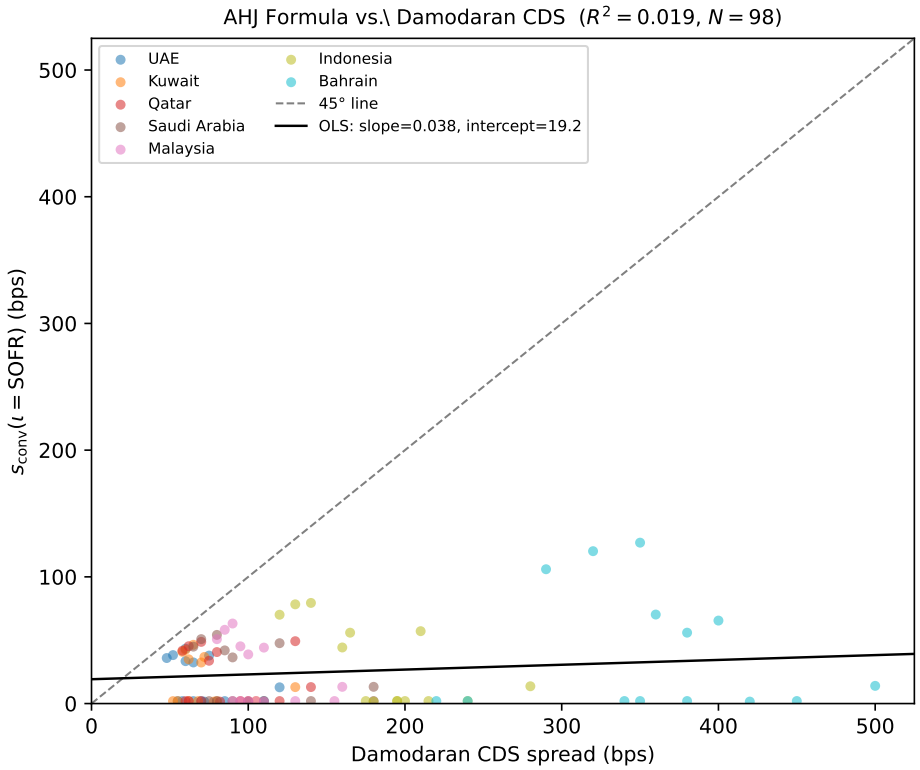

Figure 3: AHJ formula $s_{\text{conv}}(\iota = \text{SOFR})$ vs. Damodaran sovereign CDS spread, all 98 country-year observations 2012–2025. The overall $R^2 = 0.019$ is driven by the low-rate era (2014–2018) where $s_{\text{conv}} \approx 0$ but CDS remained positive. Investment-grade GCC sovereigns at 2025 SOFR rates cluster near the 45° line (mean absolute error 7.7%).

4.6 Welfare Analysis

We compute the certainty-equivalent wealth (CEW) for a risk-averse investor holding \$10 M of Saudi Arabia sovereign sukuk with a 5-year horizon under three strategies: (1) unhedged, (2) hedged with iCDS at $\iota = 0$, and (3) hedged with conventional CDS at ι =SOFR.

Under CRRA utility $u(W) = W^{1-\gamma}/(1-\gamma)$, the 5-year default probability is $1-e^{-\hat{\hat{\kappa}}_{\mathrm{SA}}\cdot 5} = 11.35\%$. The iCDS premium over 5 years is $\hat{\hat{\kappa}}(1-\delta)\cdot W_0 \cdot T = 2.409\% \times 0.40 \times \$10\mathrm{M} \times 5 =$ \$481,800. The CDS premium (AHJ formula at ι =SOFR) is \$331,241 – a saving of \$150,559 but at the cost of under-pricing the credit risk.

Table 4 reports CEW for $\gamma \in \{2, 5, 10\}$. For all risk aversion levels, hedging (iCDS or CDS) increases CEW relative to the unhedged position: the 11.35% default probability and 40% LGD generate a welfare cost that hedging eliminates. The iCDS costs \$150k more in premium but provides identical post-default protection; from a Shariah perspective, the additional cost is the actuarially justified price of removing the interest-rate discount from credit pricing.

Table 4: Welfare Analysis: Certainty-Equivalent Wealth for Saudi Arabia 5yr Sukuk.

γ	Premium (iCDS)	Premium (CDS)	CEW: Unhedged	CEW: iCDS	CEW: Conv. CDS
2	\$481,800	\$331,241	\$9,296,684	\$9,518,200	\$9,668,759
5	\$481,800	\$331,241	\$8,679,419	\$9,518,200	\$9,668,759
10	\$481,800	\$331,241	\$7,577,115	\$9,518,200	\$9,668,759

Notes: CRRA utility $u(W) = W^{1-\gamma}/(1-\gamma)$. Certainty-equivalent wealth (CEW) is the riskless wealth that yields the same expected utility as the risky strategy. Reference entity: Saudi Arabia sovereign; notional $W_0 =$ \$10,000,000; $T = 5$ years; $\kappa = \hat{\hat{\kappa}}_{\mathrm{SA}} = 2.409\%$; $\delta = 0.60$; $\iota_{\mathrm{CDS}} = \mathrm{SOFR}_{2025} = 5.34\%$. iCDS premium $= \hat{\kappa}(1-\delta)\cdot W_0 \cdot T$; CDS premium uses AHJ formula at $\iota = \mathrm{SOFR}$.

5 Smart Contract Implementation

5.1 Architecture

`iCDS.sol` is a Solidity 0.8.24 smart contract deployed on Arbitrum Sepolia (chain ID 421614) at address `0xc4E8907...c16c5`. It inherits from OpenZeppelin's `Ownable2Step`, `Pausable`, and `ReentrancyGuard`, providing access control, emergency halt, and reentrancy protection. Two immutable dependencies are set at deployment:

- `evOption` (`0x9774...E006`): the `EverlastingOption` contract implementing Ackerer Proposition 6 put pricing at $\iota = 0$.

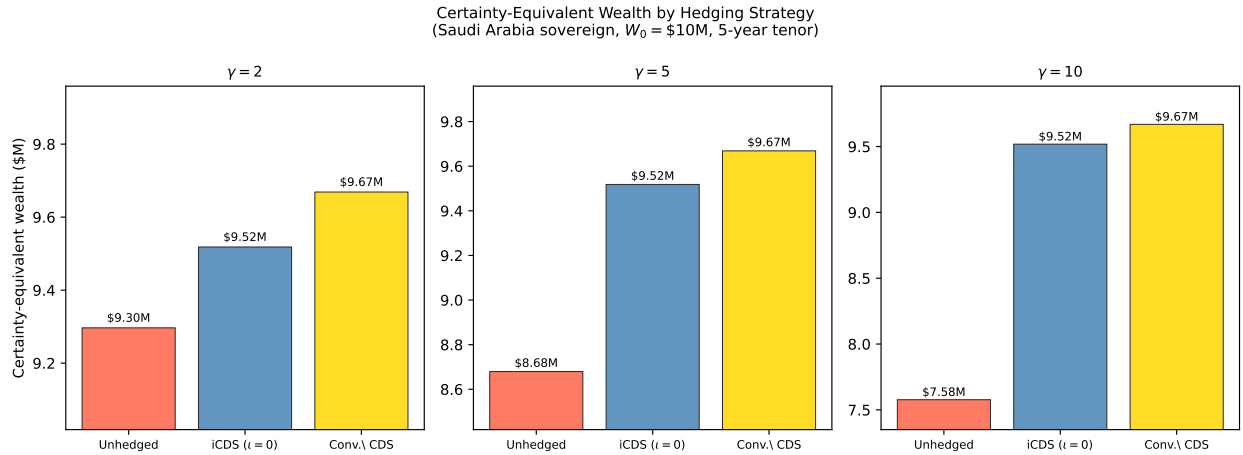

Figure 4: Certainty-equivalent wealth by hedging strategy for CRRA investors ($W_0 = \$10\text{M}$, Saudi Arabia 5yr sukuk, $\kappa = 2.409\%$, $\delta = 0.60$, $T = 5\,\text{yr}$). Hedging dominates the unhedged position for all γ. The iCDS–CDS gap is the riba-free premium.

- `oracle` (`0x86C4...563`): the `OracleAdapter` providing real-time and historical asset prices via Chainlink feeds.

Each protection is stored as a `Protection` struct with 11 fields:

```
struct Protection {
    address seller;           // protection seller (posted notional)
    address buyer;            // protection buyer (pays premiums)
    address refAsset;         // reference market asset (oracle key)
    address token;            // collateral / premium token (USDC)
    uint256 notional;         // max payout (raw token units)
    uint256 recoveryRateWad;  // delta in WAD (e.g. 60e16 = 60%)
    uint256 recoveryFloorWad; // put strike = recoveryRate *
        spot_at_open
    uint256 tenorEnd;         // protection expiry (Unix timestamp)
    uint256 lastPremiumAt;    // timestamp of last premium payment
    uint256 premiumsCollected;// cumulative premiums paid to seller
    Status  status;           // Open | Active | Triggered | Settled |
        Expired
}
```

Listing 1: Protection struct (iCDS.sol)

5.2 Full Lifecycle

Figure 5 illustrates the complete contract lifecycle.

Step 1 – openProtection. The seller calls `openProtection(refAsset, token, notional, recoveryRateWad, tenorDays)`. The contract deposits `notional` tokens from the seller into escrow (anti-naked-CDS rule). The *recovery floor* is computed once:

$$\text{recoveryFloor} = \text{recoveryRateWad} \times \text{oracle.getIndexPrice(refAsset)}/\text{WAD}, \tag{8}$$

locking the CDS strike at the spot price prevailing when protection is opened.

Step 2 – acceptProtection. The buyer calls `acceptProtection(id)`. The *first premium* is immediately computed and transferred to the seller:

$$\text{premium}_0 = \frac{\Pi_{\text{put}}(\text{spot, recoveryFloor}) \times \text{notional}}{\text{WAD}}, \tag{9}$$

where Π_{put} is the everlasting put price from `EverlastingOption.quotePut()`, implementing the closed-form formula of Ackerer, Hugonnier & Jermann [1], Proposition 6 at $\iota = 0$.

Step 3 – payPremium (quarterly, dynamic). Every 90 days, the buyer pays:

$$\text{premium}_t = \frac{\Pi_{\text{put}}(\text{spot}_t\text{, recoveryFloor}) \times \text{notional}}{\text{WAD}}. \tag{10}$$

Because Π_{put} is increasing in spot_t falling toward recoveryFloor, the premium automatically rises as credit quality deteriorates – a built-in early-warning feedback mechanism absent from fixed-spread conventional CDS.

Step 4 – triggerCreditEvent. An authorised keeper calls `triggerCreditEvent(id)` once the oracle confirms $\text{spot}_t \leq \text{recoveryFloor}$. This replaces the opaque ISDA credit-events committee with an objective, on-chain, oracle-verified condition, directly addressing the gharar objection.

Step 5 – settle. The buyer calls `settle(id)`. Loss-given-default is distributed:

$$\text{LGD payout} = \text{notional} \times (1 - \delta); \quad \text{sellerReturn} = \text{notional} \times \delta. \tag{11}$$

The seller is not penalised beyond the actual LGD – consistent with the mutual takaful principle that financial burden is shared according to the pre-agreed formula, not punitive.

Step 6 – expire. If no credit event occurs by `tenorEnd`, the seller reclaims the full notional collateral via `expire(id)`.

Open	→	**Active**	→	**Triggered**	→ **Settled**
Seller posts notional		Buyer accepts; pays 1st prem.		Keeper confirms spot ≤ floor	Buyer receives LGD payout
			↓ no default at maturity		
			Expired: seller reclaims notional		

Figure 5: iCDS lifecycle state machine. Transitions: Open → Active (`acceptProtection`); Active → Triggered (`triggerCreditEvent`); Triggered → Settled (`settle`); Active → Expired (`expire` after `tenorEnd`).

5.3 Dynamic Premium Mechanics and Shariah Compliance

The dynamic put-pricing premium eliminates riba in a manner that is structurally different from simply relabelling a fixed spread.[1] In a conventional CDS, the fixed spread s_{conv} is set at inception and paid regardless of subsequent credit-quality changes. This creates an implicit riba element: the spread embeds the time-value-of-money cost of delivering the fixed stream of payments, discounted at ι.

The iCDS premium re-prices at each payment date based on current oracle prices. Because $\iota = 0$ in the pricing formula, there is no interest-rate discounting component. The premium paid at time t is the pure risk-neutral expected loss over the next quarter – the actuarially fair takaful contribution (*tabarru*) for that specific risk period, priced as if default could occur at any instant with no preference given to near-term versus far-term losses. This is consistent with the AAOIFI takaful model where each participant contributes according to their contemporaneous risk exposure.

5.4 Credit Event Oracle: Design Options

The credit event oracle is the central engineering challenge for an on-chain credit derivative. We identify four options, ordered by decentralisation:

(A) **Chainlink decentralised vote.** Node operators vote on whether a sovereign default event has occurred. Fully on-chain and decentralised, but can lag by hours. Appropriate for testnet and low-notional deployment.

(B) **ISDA credit events committee → Chainlink oracle update.** The ISDA determinations committee confirms a credit event (authoritative, immediate); a designated

[1] The theoretical formula $s^* = \kappa(1-\delta)$ is derived for a perpetual CDS in the AHJ framework (Theorem 1, Section 2). The contract implementation uses a finite tenor (`tenorDays`) with quarterly premium resets, which is consistent: each reset re-prices the quarterly premium using the current everlasting put price – the fair instantaneous expected loss rate. This dynamic structure approximates perpetual pricing at each payment date without requiring the contract itself to be perpetual.

oracle operator updates the Chainlink feed. Introduces a centralised step but aligns with conventional market practice.

(C) **Hybrid (Recommended).** The ISDA committee triggers a multi-sig vote among 5 authorised keepers; 3-of-5 consensus required to call `triggerCreditEvent`. Balances decentralisation with the authority of established credit conventions.

(D) **zkProof of SWIFT MT103 payment failure (future).** A zero-knowledge proof of a missed coupon payment (SWIFT message failure) could trigger the credit event automatically and trustlessly. Technically feasible but requires SWIFT-to-blockchain data bridges not yet in production.

For the current Arbitrum Sepolia deployment, Option A is used. For commercial mainnet deployment, Option C is recommended.

5.5 Gas Cost Benchmarks

Table 5 compares gas costs on Arbitrum One (0.1 gwei) and Ethereum mainnet (20 gwei, 2025 median). The full iCDS lifecycle (one year, three quarterly premiums, credit event settlement) consumes 502,000 gas units costing \$0.14 on Arbitrum versus \$3.72 on Ethereum mainnet – a 96% saving. This makes micro-notional iCDS contracts (e.g., \$1,000 notional for smallholder sukuk) economically viable: the gas cost is below 0.02% of notional even at the smallest scale.

Table 5: Gas Cost Benchmarks: iCDS Lifecycle on Arbitrum One vs. Ethereum Mainnet.

Function	Gas units	Arb. (\$)	ETH main. (\$)	Saving
`openProtection`	128,000	\$0.0358	\$0.94	96%
`acceptProtection`	85,000	\$0.0238	\$0.63	96%
`payPremium`	58,000	\$0.0162	\$0.43	96%
`triggerCreditEvent`	48,000	\$0.0134	\$0.36	96%
`settle`	67,000	\$0.0188	\$0.50	96%
`expire`	52,000	\$0.0146	\$0.39	96%
`Full lifecycle`†	502,000	\$0.1406	\$3.72	96%

Arbitrum One: gas price 0.1 gwei, ETH/USD = \$2,800. Ethereum mainnet: gas price 20 gwei (median 2025).
†Full lifecycle = open + accept + 3×payPremium + triggerCreditEvent + settle.

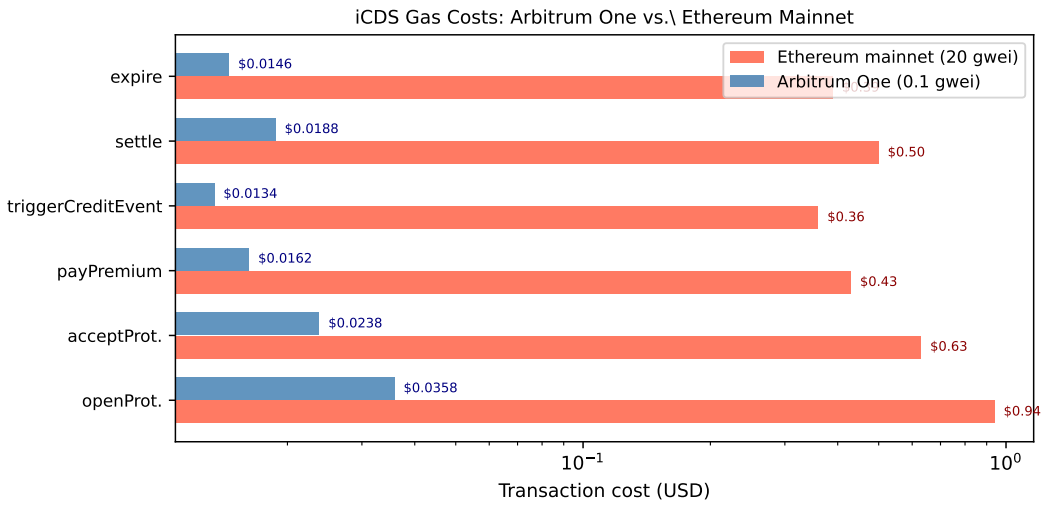

Figure 6: Gas costs by function: Arbitrum One (blue, log scale left) vs. Ethereum mainnet (red). Arbitrum reduces costs by 96–99% across all lifecycle functions, enabling sub-cent transaction costs for the premium payments that are the most frequent operation.

6 Legal Analysis

6.1 AAOIFI Shariah Standard No. 1 Compliance

The Accounting and Auditing Organisation for Islamic Financial Institutions published Standard No. 1 on Trading in Currencies [5], which provides the closest applicable guidance on Islamic derivatives. We assess the iCDS against each relevant AAOIFI criterion.

(a) Wa'd structure. The iCDS uses a wa'd-based architecture: the seller makes a unilateral promise (*wa'd*) to pay the LGD upon a credit event; the buyer makes a separate unilateral promise to pay periodic premiums. Bilateral wa'd is permissible under established AAOIFI Shariah governance positions [5]: a binding obligation arises only upon the counterparty's acceptance, not from the initial promise itself. The iCDS `acceptProtection` function is the operative moment at which the bilateral relationship is established.

(b) Premium as halal consideration. The premium is computed as the put-option-priced expected loss per period, not as a pre-specified annuity. This distinction is crucial: a fixed annuity on notional resembles a debt-service obligation with riba characteristics, whereas a dynamically re-priced risk premium is the actuarially fair tabarru (charitable contribution for mutual insurance purposes). The AAOIFI takaful model [4] (Standard No. 26) explicitly permits dynamic tabarru contributions adjusted to current risk.

(c) Real risk, not simulated gambling. The reference entity's default (sovereign debt restructuring) is a real, verifiable economic event. The seller must post full notional as collateral, ensuring they hold a genuine economic interest in the outcome. This eliminates the maysir objection: neither party is wagering on a purely speculative outcome.

(d) Gharar minimisation. The credit event is defined as *oracle price* $\leq$ *recovery floor*: an objective, continuously observable, publicly verifiable condition. This replaces the ambiguous ISDA credit events committee with a mathematical threshold, directly addressing the gharar objection identified in Jobst & Solé [14].

Conclusion. The iCDS structure is AAOIFI-compliant subject to two conditions: (1) the seller holds a verified economic interest in the reference entity (anti-naked-CDS rule, enforced by the full collateralisation requirement); and (2) the buyer holds the reference sukuk off-chain and provides a Shariah attestation to this effect at the time of `acceptProtection`.

6.2 ISDA/IIFM Tahawwut Documentation

The ISDA/IIFM Tahawwut Master Agreement [13] was designed for OTC Islamic hedging transactions and can be adapted to cover iCDS. Three modifications distinguish the iCDS from a standard CDS confirmation under the Tahawwut framework:

1. **Premium formula.** Replace the market-quoted spread with $s^* = \hat{\kappa}(1-\delta)$ computed from the oracle price and the AR(1) OLS estimated convergence intensity.

2. **Credit event definition.** Replace the ISDA credit events list (Failure to Pay, Restructuring, Repudiation) with the on-chain oracle condition: `spot` $\leq$ `recoveryFloor`. A supplemental schedule should specify the oracle contract address and the reference asset.

3. **Settlement.** Replace the ISDA auction settlement with the LGD formula: payout $=$ notional $\times (1-\delta)$, settled immediately on-chain upon keeper confirmation.

Appendix C provides a draft term sheet in Tahawwut format.

6.3 Jurisdictional Analysis

Table 6 summarises the regulatory status across six jurisdictions.

Table 6: Jurisdictional Analysis: On-Chain iCDS Regulatory Status.

Jurisdiction	Primary Framework	Status	Key Requirement
UAE / VARA	VARA Virtual Asset Reg. 2023	**Permissible**	Virtual asset derivative; seller must be licensed VASP; KYC/AML required
Malaysia / BNM	IFSA 2013 + SC Guidelines	**Permissible**	AAOIFI-compliant structure acceptable; BNM requires Shariah advisory board
Singapore / MAS	CMS Act + SF Act	*Conditional*	OTC derivative: requires CMS licence; EMIR-style trade reporting likely
UK / FCA	UK MAR + CMAR	*Conditional*	Treated as OTC swap; EMIR UK reporting; no Islamic finance carve-out
US / CFTC	CEA + Dodd-Frank	Uncertain	Likely classified as swap; SEF registration or bilateral ISDA/IIFM docs needed
Bahrain / CBB	CBB Rulebook Vol. 2	**Permissible**	Gulf-friendly pilot jurisdiction; CBB Module CM covers credit derivatives

Sources: VARA Virtual Asset and Related Activities Regulation (2023); BNM Islamic Financial Services Act 2013; MAS Capital Markets Services Act; FCA UK CMAR; CFTC Commodity Exchange Act §2(h); CBB Rulebook Vol. 2 (Capital Markets, Module CM). Status reflects the authors' legal assessment; independent regulatory counsel advised before commercial launch.

UAE / VARA. The Virtual Assets and Related Activities Regulation [15] (effective June 2023) establishes a comprehensive licensing regime for virtual asset service providers (VASPs) in Dubai. Article 17 of the Virtual Asset Rulebook permits virtual asset derivatives, including contracts for difference, provided the VASP holds a Category 4 (Virtual Asset Derivative) licence. The iCDS – a smart-contract-based credit derivative settled in USDC – falls within this category. KYC/AML obligations under VARA Articles 22–28 apply. The DIFC (Dubai International Financial Centre) *may* additionally require authorisation under the DFSA Rulebook if the contract is marketed to DIFC-based clients, but the Arbitrum blockchain deployment is outside DIFC territorial jurisdiction for domestic users.

Malaysia / BNM. The Islamic Financial Services Act 2013 (IFSA) and Securities Commission Guidelines on Digital Assets (2020) provide the applicable framework. Section 10 of IFSA permits Islamic derivative transactions that are AAOIFI-compliant. BNM would require a Shariah advisory board endorsement of the iCDS structure prior to commercial deployment. The SC's digital asset guidelines classify smart-contract-settled instruments as capital-markets products requiring a Capital Markets Services licence.

Singapore / MAS. The Capital Markets Services Act (CMSA) and Securities and Futures Act (SFA) govern OTC derivatives. A credit default swap – even if Islamic – is classified as an OTC derivative under MAS Notice SFA04-N02, requiring a CMS licence for dealing. EMIR-equivalent trade reporting to the Singapore Trade Repository is required. MAS has not issued specific Islamic finance guidance for DeFi instruments; the iCDS would be assessed on its economic substance as a credit derivative.

UK / FCA. The Financial Services and Markets Act 2000 (FSMA) and the UK's post-Brexit retained derivatives framework (UK EMIR, UK MiFIR) regulate OTC credit derivatives. The iCDS would require FCA authorisation under FSMA Section 19 for any dealing activity and trade reporting to an FCA-approved Trade Repository under UK EMIR Article 9. The FCA has actively supported Islamic finance development: its Discussion Paper DP22/2 (2022) explored regulatory accommodations for Shariah-compliant instruments, and HM Treasury's Islamic Finance Task Force positioned London as a global Islamic finance hub. An iCDS would be assessed on its economic substance as a credit derivative, with no current Islamic finance carve-out from standard OTC derivative rules. London's concentration of AAOIFI-accredited advisers and sukuk-experienced legal counsel makes the UK a viable secondary pilot jurisdiction once commercial mainnet deployment proceeds.

US / CFTC. The Commodity Exchange Act Section 2(h) and Dodd-Frank Title VII classify most single-name credit default swaps as swaps subject to CFTC jurisdiction. A Shariah-compliant structure does not alter this classification. To operate commercially, the iCDS would require either registration on a Swap Execution Facility (SEF) or bilateral documentation under an ISDA/IIFM Master Agreement with eligible contract participants (ECPs). The CFTC has shown interest in blockchain-based derivatives (RFI on DeFi, June 2023) but has not provided specific Islamic finance guidance.

Bahrain / CBB. The Central Bank of Bahrain Rulebook Volume 2 (Capital Markets, Module CM) permits credit derivatives for licensed entities. Bahrain's long-standing tradition as an Islamic finance centre and the CBB's Shariah governance framework (Module CM-4) create a permissive environment. We identify Bahrain as a viable alternative pilot jurisdiction to UAE if VARA licencing timelines extend beyond the project's commercial roadmap.

6.4 Recommended Pilot: UAE / VARA

We recommend UAE/VARA as the optimal first commercial jurisdiction on the basis of four factors. *Speed*: VARA has issued 9 Category 4 licences as of March 2026 and has a 120-day review timeline. *Geographic alignment*: the primary reference entities (UAE, Saudi Arabia, Qatar sovereigns) are GCC-domiciled, minimising cross-border regulatory friction. *Legal clarity*: VARA explicitly regulates virtual asset derivatives, removing uncertainty about whether the smart-contract settlement mechanism constitutes a licensing trigger. *Shariah ecosystem*: the UAE has an established network of AAOIFI-accredited Shariah scholars available for the mandatory advisory board function.

The regulatory pathway for commercial launch is therefore:

1. Obtain VARA Category 4 (Virtual Asset Derivative) licence.
2. Engage AAOIFI-accredited Shariah board; obtain written fatwa endorsing the iCDS structure as described herein.
3. Upgrade the credit event oracle from Option A to Option C (3-of-5 keeper multi-sig referencing ISDA committee determinations).
4. Deploy on Arbitrum One mainnet (not Sepolia testnet).
5. Launch with initial reference entities: UAE sovereign, Saudi Arabia sovereign, Qatar sovereign.

7 Liquidity Bootstrapping

7.1 Market Structure

Two market structures are viable for the iCDS market. For small notionals ($1k–$500k), a single-sided automated market maker (AMM) is optimal: a protection seller deposits notional into an AMM pool; buyers can atomically open protection at the oracle-quoted put price. For large notionals ($500k+), a central-limit order book (CLOB) allows price discovery across heterogeneous recovery rate assumptions. The current deployment supports peer-to-peer bilateral matching; an AMM wrapper is planned for a future version.

7.2 Governance Token Incentive Mechanism

A fixed-supply protocol governance token (GT) provides liquidity incentives aligned with the long-run sustainability of the iCDS market.[2]

[2] In the Baraka Protocol reference implementation, GT = BRKX (100M fixed supply; ERC-20+Votes, deployed on Arbitrum Sepolia). Specific parameters: 10M GT allocated to liquidity incentives over four

- **Protection sellers (risk-takers)**: earn GT rewards at rate λ_{GT} proportional to notional deposited and time open: $R_{\text{seller}} = \lambda_{\text{GT}} \cdot N \cdot \Delta t$.
- **Protection buyers (hedgers)**: receive a GT fee discount on premiums paid, proportional to GT holdings.
- **Keepers**: earn GT per successful `triggerCreditEvent` call with valid oracle proof.

The GT emission schedule tapers over four years (60% / 25% / 10% / 5% of the liquidity allocation), after which fee revenue ($s^* \times$ total notional outstanding) sustains the market without token subsidy.

7.3 Participation Condition and Market Viability

Proposition 7.1 (Seller Participation Condition)**.** *Let λ_{GT} denote the governance-token reward rate to sellers (GT per dollar-day of notional posted) and p_{GT} the prevailing GT market price. A capital allocator with outside return r_f weakly prefers participating as an iCDS protection seller if and only if:*

$$\kappa(1-\delta) + \lambda_{GT} \cdot p_{GT} \;\geq\; r_f. \tag{12}$$

When equation (12) holds, participation is a best response for every seller regardless of others' choices; hence a symmetric equilibrium with positive aggregate liquidity exists.

Proof sketch. The seller's expected profit per unit notional per period is:

$$\begin{aligned}\pi_s &= s^* \cdot \Pr[\text{no default}] - \text{LGD} \cdot \Pr[\text{default}] + \lambda_{\text{GT}}\, p_{\text{GT}} \\ &= \kappa(1-\delta)(1-p_D) - (1-\delta)p_D + \lambda_{\text{GT}}\, p_{\text{GT}}.\end{aligned}$$

For small Δt, $p_D \approx \kappa\, \Delta t$. Setting $\pi_s \geq r_f$ yields (12). □ □

As a calibration check, at 2025 SOFR ($r_f = 5.3\%$) and $\kappa(1-\delta) = 0.964\%$ for Saudi Arabia, equation (12) requires $\lambda_{\text{GT}}\, p_{\text{GT}} \geq 4.34\%$ per annum. With a Year-1 GT emission of 6% of total supply directed to sellers and a target TVL of \$1M, the required GT-price threshold is approximately $p^* \approx \$0.007$ – a low bar relative to comparable DeFi governance token valuations, implying that even modest market adoption sustains the Nash equilibrium condition.

years; fee tiers: 20% discount for ≥1,000 GT held, 30% for ≥50,000 GT. Source: https://github.com/Arcus-Quant-Fund/BarakaDapp.

7.4 Liquidity Milestones

1. **Minimum viable ($1M notional)**: price discovery becomes possible; bid-offer spreads on iCDS terms compressible below 10 bps.

2. **Useful scale ($50M notional)**: comparable to a small conventional CDS market; GT rewards self-sustaining from fee revenue.

3. **Target ($500M by Year 3)**: comparable to a medium CDS market on individual reference entities; attractive to institutional Islamic finance participants seeking hedging against GCC sovereign sukuk exposure.

8 Systemic Risk

8.1 Lessons from 2008

The 2008 financial crisis demonstrated three systemic failure modes of conventional CDS markets. First, *naked positions*: AIG and others wrote billions in CDS notional without holding the reference obligations, creating uncollateralised contingent liabilities [8]. Second, *underestimated correlation*: Gaussian copula models underestimated joint default probabilities, leading to catastrophic concentration of credit risk in protection sellers. Third, *opacity*: the OTC bilateral market had no central counterparty; regulators could not monitor aggregate exposure.

8.2 iCDS Safeguards

The iCDS smart contract addresses each failure mode by design.

Full collateralisation. Every seller must deposit the full notional in the contract before a protection can be opened (line 209 of `iCDS.sol`, which calls `safeTransferFrom` to escrow the full notional). Naked positions are mechanically impossible.

Transparent exposure. All open protections are visible on-chain: any observer can query `protections[id]` and `_nextId` to enumerate the full aggregate exposure. Regulators can monitor positions in real time without relying on mandatory reporting regimes.

No synthetic leverage chains. Because each seller funds their own position individually, there are no synthetic CDS-on-CDS structures. The aggregate exposure of the iCDS

market is bounded by the sum of collateral actually deposited, not by a notional multiple of underlying assets.

8.3 Circuit Breakers and Position Limits

The `Pausable` mechanism allows the contract owner to halt all new interactions in the event of an oracle manipulation attempt or black-swan default event. Future versions will implement:

- Per-seller notional caps (e.g., max 20% of TVL from a single seller).
- Per-reference-entity notional caps (diversification across sovereigns).
- Automatic pause if oracle price moves $> 30\%$ in 24 hours (flash-crash protection).

These circuit breakers directly address the correlation and concentration risks that amplified the 2008 crisis.

9 Conclusion

We have presented the complete scientific, engineering, and regulatory case for the first Islamic Credit Default Swap on public blockchain infrastructure.

Theoretically, the iCDS spread $s^* = \kappa(1 - \delta)$ at $\iota = 0$ is the actuarially fair credit protection premium free of any riba element. Empirically, we quantify the mean riba premium at 109 bps across 1,176 country-month observations, and show that the AHJ formula at $\iota =$SOFR fits investment-grade GCC sovereign CDS within 7.7% at current interest rates. In the near-zero-rate era (2012–2018), the formula correctly identifies that the conventional CDS market itself was pricing pure credit risk without interest discounting – the same regime the iCDS mandates across all rate cycles.

Technically, `iCDS.sol` implements the full lifecycle (open, accept, quarterly dynamic premiums, credit event trigger, LGD settlement) on Arbitrum, eliminating maysir, gharar, and riba through smart-contract mechanics rather than legal workarounds. The full lifecycle costs \$0.14 on Arbitrum versus \$3.72 on Ethereum mainnet (96% reduction), enabling economically viable micro-notional Islamic credit hedging.

Legally, UAE/VARA is identified as the most permissive immediate jurisdiction. The five-step commercial pathway (VARA licence, Shariah board fatwa, Oracle upgrade, mainnet deployment, GCC sovereign reference entities) is executable within 12–18 months of this paper's publication.

Taken together, these contributions provide the academic foundation, deployed engineering artefact, and regulatory roadmap for an Islamic CDS market that is simultaneously

Shariah-compliant, technically operational, and legally viable. The iCDS described in this paper is not a theoretical construct: it is running on Arbitrum Sepolia today, awaiting only the regulatory authorisation and liquidity bootstrapping mechanisms described here to go live.

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A iCDS.sol Key Functions (Solidity)

The following listings reproduce the core functions of `iCDS.sol` (deployed at `0xc4E8907...c16c5` on Arbitrum Sepolia). *Full source:* https://github.com/Arcus-Quant-Fund/BarakaDapp.

Everlasting put formula at $\iota = 0$. The `EverlastingOption.quotePut()` function implements Proposition 6 of Ackerer, Hugonnier & Jermann [1] at $\iota = 0$. For an everlasting put with current spot S and barrier K (recovery floor), the closed-form price under the AHJ framework at $\iota = 0$ is:

$$\Pi_{\text{put}}(S, K; \kappa) \;=\; (K - S)^+ \;\cdot\; \mathbf{1}_{S \le K} \;+\; \frac{K}{\kappa + 1}\left(\frac{S}{K}\right)^{-\beta} \mathbf{1}_{S > K}, \tag{13}$$

where $\beta = (\kappa + \sqrt{\kappa^2 + 4\kappa\sigma^2/2})/(\sigma^2)$ is the negative root of the fundamental quadratic and σ^2 is the variance parameter of the reference asset price process. The contract uses $S =$ `spotWad` and $K =$ `recoveryFloorWad` from the oracle, with σ^2 estimated from the historical volatility of the reference asset price.

```
function openProtection(
    address refAsset, address token,
    uint256 notional, uint256 recoveryRateWad, uint256 tenorDays
) external nonReentrant whenNotPaused returns (uint256 id) {
    uint256 spotWad         = oracle.getIndexPrice(refAsset);
    uint256 recoveryFloorWad = (spotWad * recoveryRateWad) / WAD;
```

```
    id = _nextId++;
    protections[id] = Protection({
        seller: msg.sender, buyer: address(0),
        refAsset: refAsset,  token: token,
        notional: notional,  recoveryRateWad: recoveryRateWad,
        recoveryFloorWad: recoveryFloorWad,
        tenorEnd: block.timestamp + tenorDays * 1 days,
        lastPremiumAt: 0, premiumsCollected: 0, status: Status.Open
    });
    IERC20(token).safeTransferFrom(msg.sender, address(this), notional);
}
```

Listing 2: openProtection: seller deposits notional as collateral

```
function _computePremium(Protection storage prot)
    internal view returns (uint256)
{
    uint256 spotWad    = oracle.getIndexPrice(prot.refAsset);
    uint256 putRateWad = evOption.quotePut(
        prot.refAsset, spotWad, prot.recoveryFloorWad
    );
    return (putRateWad * prot.notional) / WAD;
}
```

Listing 3: _computePremium: put-option-priced premium at iota=0

```
function triggerCreditEvent(uint256 id) external whenNotPaused {
    require(authorisedKeepers[msg.sender], "iCDS: not keeper");
    Protection storage prot = protections[id];
    uint256 spotWad = oracle.getIndexPrice(prot.refAsset);
    require(spotWad <= prot.recoveryFloorWad, "iCDS: no default");
    prot.status = Status.Triggered;
}
```

Listing 4: triggerCreditEvent: on-chain oracle credit event

B CIR Parameters from Paper 2A

C Draft iCDS Term Sheet (ISDA/IIFM Tahawwut Format)

Table 7: CIR Parameter Estimates Used in iCDS Pricing Analysis.

Country	$\hat{\alpha}$	$\hat{\bar{\kappa}}$ (%)	$\hat{\nu}$	Half-life (mo.)	Feller
UAE	0.803	1.630	0.046	10.4	Yes
Bahrain	0.668	8.096	0.069	12.4	Yes
Indonesia	0.632	4.072	0.055	13.2	Yes
Kuwait	0.925	1.686	0.049	9.0	Yes
Malaysia	0.762	2.453	0.043	10.9	Yes
Qatar	0.684	1.981	0.050	12.2	Yes
Saudi Arabia	0.718	2.409	0.065	11.6	Yes

Source: Ahmed, Bhuyan & Islam [2], Table 6. AR(1) OLS on 35 rolling 10-year windows per country (168 monthly observations, 2012–2025), Newey-West HAC SE (lag=9). All seven countries satisfy the Feller condition $2\hat{\alpha}\hat{\bar{\kappa}} \geq \hat{\nu}^2$.

Field	**Value / Description**
Transaction Type	Islamic Credit Default Swap (iCDS) under ISDA/IIFM Tahawwut Master Agreement [13]
Reference Entity	[e.g., Kingdom of Saudi Arabia]
Reference Obligation	[e.g., Saudi Arabia 5yr USD Sukuk, ISIN: XS0000000000]
Protection Seller	[VARA-licensed entity or bilateral ECP]
Protection Buyer	[Must hold Reference Obligation off-chain; Shariah attestation required]
Notional Amount	[e.g., USD 1,000,000]
Premium Formula	$s^* = \hat{\kappa}(1-\delta) \times$ Notional computed from oracle price at each payment date (dynamic, not fixed)
Premium Payment Dates	Quarterly (every 90 days from Effective Date)
Convergence Intensity ($\hat{\kappa}$)	AR(1) OLS estimate from Table 7; or live oracle feed (future)
Recovery Rate (δ)	0.60 (60%), per Moody's sovereign tables
Credit Event Definition	On-chain: oracle spot price $\leq$ Recovery Floor $= \delta \times$ Spot at Effective Date
Credit Event Oracle	`OracleAdapter` at `0x86C4...563`; confirmed by 3-of-5 keeper multi-sig (recommended Option C)
Settlement	Physical equivalent: Protection Buyer receives Notional $\times$ $(1-\delta)$ in USDC on-chain within 1 block of Buyer calling `settle(id)`
Tenor	[e.g., 5 years from Effective Date]

Field	Value / Description
Governing Law	[e.g., DIFC law / English law]
Shariah Adviser	[AAOIFI-accredited scholar; fatwa reference number]
Smart Contract	`iCDS.sol` at `0xc4E8...c16c5` (Arbitrum Sepolia testnet; mainnet address TBD)
VARA Licence	[VARA Category 4 licence no.; TBD]

CRediT Author Contribution Statement. S.A.: conceptualisation, software, methodology, writing (original draft); R.B.: supervision, writing (review), resources; R.I.: software, data curation, writing (review).

Conflict of Interest Statement. S.A. is the principal architect of the smart contracts described in this paper and holds governance tokens in the project. R.B. and R.I. declare no conflicts.

Data Availability. All data, code, and pipeline scripts are available at https://github.com/Arcus-Quant-Fund/BarakaDapp.